# Prisha Aggarwal

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## Professional Summary

Computer Engineering student with research experience at NIT Delhi and hands-on experience in Machine Learning, Computer Vision, and software development. Skilled in Python, C++, SQL, Deep Learning, and OpenCV, with strong foundations in DSA, OOP, and problem-solving, and practical project development.

## Education

**Thapar Institute Of Engineering and Technology** **July 2023 - Present**
*B.E Computer Engineering, CGPA : 8.59* *Patiala, Punjab*

**MGN Public School** **2021 - 2023**
*Senior Secondary, XII : 90.2% —| JEE Mains : 93.96 percentile* *Jalandhar, Punjab*

**Cambridge International School For Girls** **2008 - 2021**
*High Secondary, X : 94.2%* *Jalandhar, Punjab*

## Experience

**Student Attentiveness Model — Research Intern** **June 2025 - August 2025**
*National Institute of Technology,Delhi* *Delhi*

- Built a real-time student attentiveness detection system using Python and OpenCV, extracting gaze, head-pose, and eye-openness features and training a multiclass Logistic Regression model to classify attentive, distracted, and sleepy states with 89% accuracy.

**Road Monitoring System ELC — Research Intern** **June 2024 - August 2024**
*Thapar Institute of Engineering and Technology* *Patiala, Punjab*

- Developed a V2V communication module using a YOLO-based model and Raspberry Pi to detect collisions and low-visibility, broadcasting alerts to nearby vehicles for safer driving with 98% accurate accident and traffic detection.

## Design Registration

**Secure Biometric USB Drive** **January 2025 - March 2025**

- Designed a secure USB drive incorporating fingerprint-based biometric authentication and storage-level monitoring, using AutoCAD.
- Design registered and published (Application No. 460865-001; Journal No. 16/2026).

## Projects

**FistulaNet: MRI Fistula Diagnosis** **January 2026 - Present**

- Developing a deep learning-based MRI analysis system for detecting perianal fistulas and abscesses, extracting clinical features, and predicting post-treatment healing time to support diagnosis and treatment planning. (GitHub)

**CampusPay: Banking-Based Smart Payment System** **January 2025 - May 2025**

- Designed and implemented a database-driven campus payment system using MySQL, supporting wallet management, top-ups, peer-to-peer transfers, and secure transaction tracking. (GitHub)

## Skills

**Languages**: Python, C++, C, JavaScript, HTML/CSS, SQL, Excel
**AI/ML**: Machine Learning, Deep Learning, Transformers, GANs, CNNs, RNNs, LSTMs, GRUs, Computer Vision, NLP
**Frameworks & Libraries**: TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib
**Core CS**: Data Structures Algorithms(70+ problems solved on LeetCode), Object-Oriented Programming, DBMS, OS
**Developer Tools**: Git, GitHub, VS Code
**Others**: AutoCAD, Public Speaking, Leadership, Adobe Photoshop

## Position of Responsibility

Class Representative — August 2024 - Present
Executive Member - Student Alumni Interaction Cell — January 2025 - December 2025
Overall Event Coordinator - Student Alumni Interaction Cell — January 2025 - March 2025